

Section 1

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 2

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 3

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 4

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 5

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 6

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 7

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 8

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 9

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 10

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 11

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 12

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 13

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 14

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 15

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 16

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 17

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 18

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader,

resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 19

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader,

resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 20

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader,

resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 21

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader,

resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 22

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader,

resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 23

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader,

resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 24

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 25

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 26

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an

investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 27

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 28

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 29

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 30

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 31

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve.

When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 32

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 33

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 34

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and

decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 35

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 36

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire

recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 37

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 38

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 39

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging

offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

Section 40

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual

decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it,

embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.

The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact. The distribution of classified material across multiple authorised recipients creates an attribution problem that conventional access control cannot resolve. When a document is encrypted once and decrypted independently by each holder, the resulting plaintext is byte-identical for every party. Any subsequent disclosure therefore implicates the entire recipient set equally, and server side logging offers no remedy because a privileged operator may alter those records after the fact. A forensic marking scheme must instead bind each individual decryption event to the identity that performed it, embedding that binding in the document itself rather than in an external register that can be rewritten. The mark must be imperceptible to the reader, resistant to casual removal, and verifiable by an investigator holding only the leaked artefact.